# Simon Dräger

sfdraeger at gmail.com | sdraeger | Google Scholar

## Education

**M.S. Computer Science** — Jan 2023 – Mar 2025
University of California, Davis — *Davis, CA, USA*
GPA: 3.96/4.0. Research in machine learning and multimodal perception.

**B.Sc. Computer Science** — Oct 2019 – Sep 2022
University of Düsseldorf — *Düsseldorf, Germany*
Thesis grade: 1.0, highest possible grade.

## Publications

### Journal and Conference Papers

- C. Lainscsek, P. Salami, **S. Dräger**, A. R. Bulsara, S. S. Cash, and T. J. Sejnowski. "Causal Dynamic Resonance." Accepted to the *Proceedings of the National Academy of Sciences (PNAS)*. Preprint: arXiv:2508.16733, 2025.
- X. Xiang, **S. Dräger**, and J. Zhang. "3DifFusionDet: Diffusion Model for 3D Object Detection with Robust LiDAR-Camera Fusion." *ICASSP 2026.*
- X. Xiang, **S. Dräger**, and J. Zhang. "EffiPerception: A Plug-and-Play Efficiency Enhancement Framework for 2D and 3D Perception Models." *PRICAI 2025.*

### Workshop Papers

- **S. Dräger.** "Halting in Recurrent Sequence Models." Accepted to the *NeurIPS 2026 Workshop on Interpretability as a Science.*
- **S. Dräger.** "Calibrating Latent Traveling Waves Controls Memory Retrieval in Recurrent Networks." Accepted to the *NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations (NeurReps).*
- **S. Dräger.** "Task-Restricted Symmetries in Recurrent Weight Space." *ICML 2026 Workshop on Weight-Space Symmetries: from Foundations to Practical Applications.*

### Submitted Abstracts

- **S. Dräger** and R. Gao. "Small input perturbations reveal the identity of task-trained recurrent neural networks." Submitted to *COSYNE 2027.*

### Preprints

- **S. Dräger** and J. Dunkelau. "Evaluating the Impact of Loss Function Variation in Deep Learning for Classification." Preprint: arXiv:2210.16003, 2022.

## Research Experience

**Research Software Engineer** — Oct 2026 – Present
Salk Institute for Biological Studies — *La Jolla, CA, USA*
Center for AI and Research Computing, Talmo Pereira

- Set up the Center for AI and Research Computing (AIRC) at the Salk Institute.
- Integrated AI methods, including large language models (LLMs), into scientists' research.

**Scientific Programmer** — Apr 2025 – Oct 2026
Salk Institute for Biological Studies — *La Jolla, CA, USA*
Computational Neurobiology Laboratory, Terrence Sejnowski

- Developed Schur-coordinate ablations of recurrent neural networks to identify which weight changes preserved memory and sequence generation; authored the resulting ICML 2026 WSS workshop paper.
- Coauthored Causal Dynamic Resonance, accepted to PNAS, on identifying false causal links through noise perturbations, with evaluations on coupled dynamical systems and intracranial EEG.
- Led development of DDALAB, an application for neurophysiological time-series analysis.

**Visiting Scholar** Jun 2024 – Sep 2024
Salk Institute for Biological Studies *La Jolla, CA, USA*
Computational Neurobiology Laboratory, Terrence Sejnowski
– Built and presented the initial DDALAB prototype, turning analysis scripts into an application refined through feedback from researchers.
– Mentored students in computational neurobiology, scientific programming, and data analysis.

**Graduate Student Researcher** Apr 2023 – Apr 2024
University of California, Davis *Davis, CA, USA*
Supervisor: Jiawei Zhang
– Developed training, evaluation, and ablation pipelines for 3DifFusionDet (ICASSP 2026).
– Collaborated on model development and experiments for EffiPerception (PRICAI 2025).

## Research Software

**DDALAB**
Python, Qt, Rust
– Desktop application and command-line interface for Delay Differential Analysis (DDA) of electrophysiology recordings, with BIDS integration, analysis history, and remote computation through the Neuroscience Gateway. GitHub

**DelayDifferentialAnalysis.jl and dda-py**
Julia, Python
– Packages exposing DDA to Julia and Python workflows: native CPU and CUDA-backed regression in Julia; NumPy arrays, MNE recordings, and BIDS datasets in Python. Julia Python

## Presentations

**Poster Presentation** 2024
Dynamics Days *Davis, CA, USA*
– The Importance of Architecture Choice in Deep Learning for Climate Applications.

## Service and Teaching

**Workshop Reviewer** 2026
NeurIPS 2026
– Interpretability as a Science
– Interpretability for Discovery
– Symmetry and Geometry in Neural Representations (NeurReps)
– Diffusion Language Models: Foundations, Efficiency, and Reasoning (DiffuLM)

**Workshop Organizer** Aug 2025
Delay Differential Analysis (DDA)
– Organized a virtual workshop for 25 participants on DDA methods and software. Workshop page

**Teaching Assistant** Summer 2022
University of Düsseldorf *Düsseldorf, Germany*
– Developed course materials, exams, and problem sets for undergraduate Statistical Data Analysis.

## Technical Skills

**Programming and tools:** Python, Julia, Rust, C, TypeScript; Qt, Docker, Git, SQLite, PostgreSQL.
**Scientific computing:** PyTorch, TensorFlow, scikit-learn, Hugging Face, NumPy, MNE.
**Methods:** recurrent-network analysis, controlled ablations, nonlinear time-series analysis.